EE 122: Introduction to Control Systems

Problem Set #12: Practice Final Exam

Name: ______________ **Submission Date:** ______________

Problem 1 For a system modeled as

$$\frac{dx}{dt} = ax + u$$

where a is a constant and u is the control input, answer the following questions:

(a) [5 points] What is the order of the system? How many states does it have?

(b) [5 points] Is the system stable for all values of a? If not, for which values of a is the system unstable?

(c) [10 points] Design a state feedback controller $u = -Kx$ that stabilizes the system for any value of a. What is the range of values for K that achieves this stabilization?

(d) [5 points] Derive the closed-loop system dynamics under the state feedback controller you designed in part (c). What is the new system matrix?

(e) [25 points] Repeat the above for the following system:

$$\frac{dx}{dt} = Ax + Bu$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ b_1 \end{bmatrix}.$$

Problem 2 [15 points] Solve Problem 9.8 in the textbook.

Problem 3

(a) [5 points] Sketch the pole-zero diagram of the following transfer function:

$$G(s) = K\frac{(s+5)(s+1)}{(s+3)(s^2+4s+8)}.$$

Answer the following questions based on the pole-zero diagram:

(b) [5 points] Is the system stable? Why or why not?

(c) [10 points] Sketch an approximate magnitude Bode plot of the system. Label the slopes and the corner frequencies on the plot.

(d) [15 points] We want to obtain a system that attenuates the load disturbance at low frequencies. That is, we want a gain of less than 0 dB at frequencies lower than 1 rad/s. Sketch a desirable Bode plot for the loop gain $L(s) = G(s)C(s)$ where $C(s)$ is a controller that we can design. Label the corner frequencies on the plot. You are not asked to find the controller — simply sketch a desirable Bode plot.